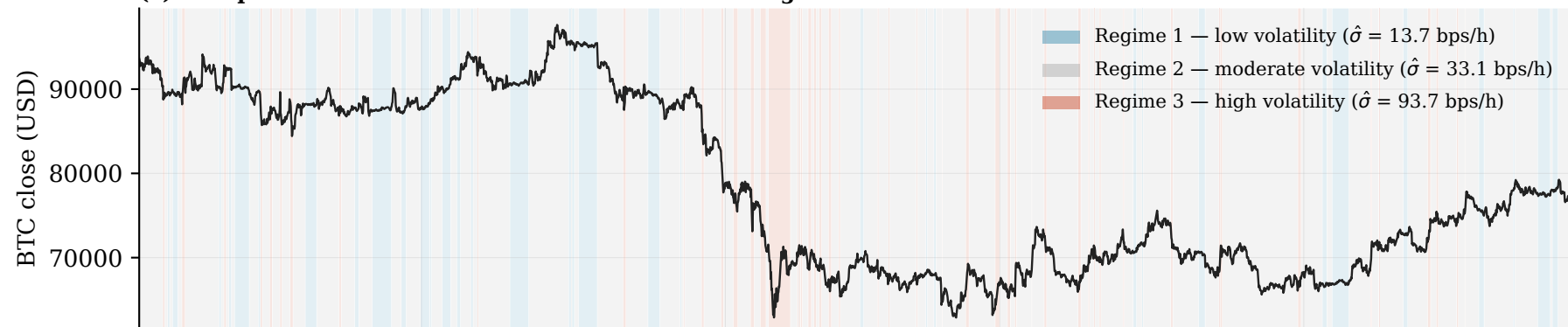


BTC perpetual on Hyperliquid — return regimes (K=3) and funding regimes

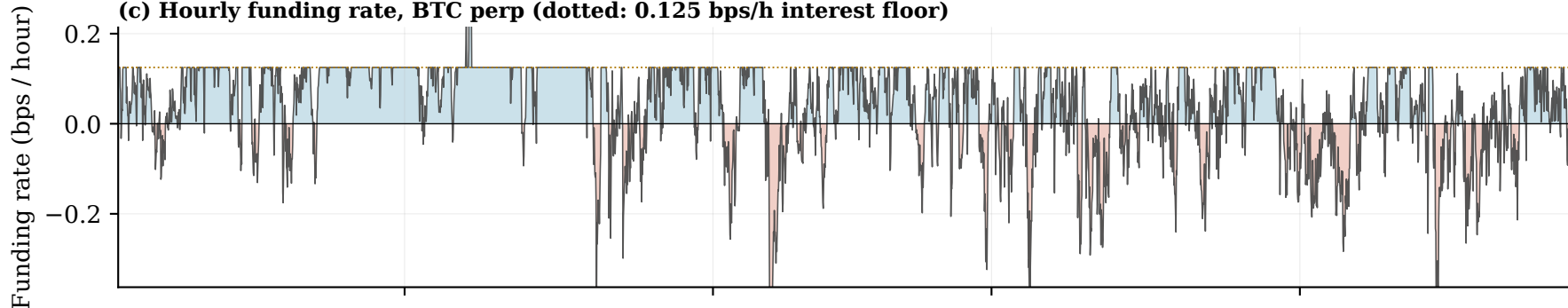
(a) BTC price with median-filtered dominant return regime



(b) Filtered posterior $P(S_t = k | \mathcal{F}_t)$ — return HMM



(c) Hourly funding rate, BTC perp (dotted: 0.125 bps/h interest floor)



(d) Filtered posterior — MS-AR(1) on premium (teal = calm, red = stressed)



(e) Realised vs. regime-implied volatility

